ZERO SIGHT

Behavioral Anomaly Detection
for Zero-Day Protocol Exploits

Network Flow Fingerprinting with Explainable AI

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19,950
Training Flows

31
CICIDS Features

17
Attack Classes

89.9%
IF F1 Score

94.4%
RF Recall (macro)

Isolation Forest · Random Forest · SHAP-Style XAI · CICIDS2017 · FastAPI

INTRODUCTION

01

The Zero-Day Problem

Signature-based IDS (Snort, Suricata) only detect attacks after a signature is written. Zero-day exploits pass through undetected during the most dangerous window hours to weeks before signatures exist.

02

Network Traffic Scale

Modern enterprise networks generate terabits per second across thousands of services. Manual inspection is operationally impossible - automated behavioral detection is the only scalable approach.

03

The Signature Lag

The gap between exploit discovery and signature deployment ranges from hours to weeks. Log4Shell (2021) was actively exploited within hours of disclosure long before any IDS had a valid signature.

04

Alert Fatigue Crisis

60%+ of IDS alerts go uninvestigated due to high false-positive rates and unexplainable detections. A system that fires without explaining why provides near-zero operational value.

05

The Explainability Gap

ML anomaly detectors produce black-box scores. Analysts cannot verify, trust, or act on a score of '87 - CRITICAL' without knowing which flow features triggered it. SHAP attribution bridges this gap.

06

ZeroSight Solution

A hybrid dual-model system combining Isolation Forest (unsupervised, zero-day capable) with Random Forest (supervised, 18-class) and SHAP-style attribution trained on synthetic CICIDS2017, validated on live PCAP captures.

SUMMARY OF ISSUES

01

Signature Systems Cannot Detect Zero-Day Attacks

Snort and Suricata match packets against known signatures. Any attack without a matching signature passes undetected. This is a fundamental architectural limitation no tuning or configuration can fix it. ZeroSight replaces signature matching with behavioral baseline deviation scoring.

02

No Explainability in Existing ML IDS Tools

Existing ML-based IDS tools (DeepLog, LogAnomaly) classify flows as normal or anomalous without feature attribution. Analysts receive a binary label with no evidence. This produces alert fatigue and undermines trust operators cannot distinguish a true positive from a false alarm.

03

False Positive Rate Unacceptable for Production Deployment

High false positive rates cause alert queues to overflow. Studies report 60%+ of IDS alerts go uninvestigated. ZeroSight's cross-model consensus (IF + RF agreement for escalation) and SHAP attribution reduce false positives and make each remaining alert explainable.

04

PCAP Analysis Requires Server Upload Privacy Violation

Most network analysis tools require uploading raw PCAP files to a cloud service. Network captures contain sensitive enterprise traffic. ZeroSight's browser-native binary PCAP parser processes captures entirely client-side - no packet data ever leaves the analyst's machine.

PROPOSED SYSTEM

01

Synthetic CICIDS2017 Dataset Generation

19,950-sample synthetic dataset: 9,750 BENIGN flows across 4 subtypes and 600 samples per 17 attack class. Flag counts proportional to packet count (ACK ~82%) eliminating a key false positive source.

02

Isolation Forest - Unsupervised Anomaly Detection

300-tree IF trained exclusively on BENIGN flows. Threshold at 90th percentile of BENIGN scores (theta = 0.50043). Any flow deviating from the learned BENIGN distribution including zero-day attacks is flagged.

03

Random Forest - 18-Class Attack Classification

100 trees, max_depth=8, class_weight=balanced. Trained on all 19,950 labeled flows with Gaussian noise augmentation (sigma=0.30). Provides specific attack labels across 18 classes with per-class confidence.

04

SHAP-Style XAI Feature Attribution

contribution_i = |scaled_value_i| × RF_importance_i. Returns top-15 features. Four rate features always guaranteed in output. Analysts see exactly which flow characteristics drove each detection decision.

05

Browser-Native Binary PCAP Parser

JavaScript PCAP parser runs entirely in the browser no server upload. Extracts real timestamps, groups bidirectional flows by canonical 5-tuple, tracks fwd/bwd packets, computes IAT from sorted timestamps.

06

FastAPI Inference Server + Interactive Dashboard

FastAPI on localhost:8000: POST /predict, POST /batch, GET /model-info, GET /health. Dashboard: threat score, severity badge, RF label, SHAP chart, deviation table, alert feed, model metrics.

SYSTEM ARCHITECTURE DIAGRAM

5-Layer Pipeline: Data Input → Feature Extraction → ML Inference → XAI Attribution → Output & API

1

Data Input

Raw PCAP Binary File

Browser JS PCAP Parser

2

Feature Extraction

31 CICIDS2017 Flow Features

StandardScaler · 31 Features

3

ML Inference

IF + RF Dual Model

IsolationForest 300 trees · RF 100 trees

4

XAI Attribution

SHAP-Style Top-15

$contribution\_i = |z\_i| \times importance\_i$

5

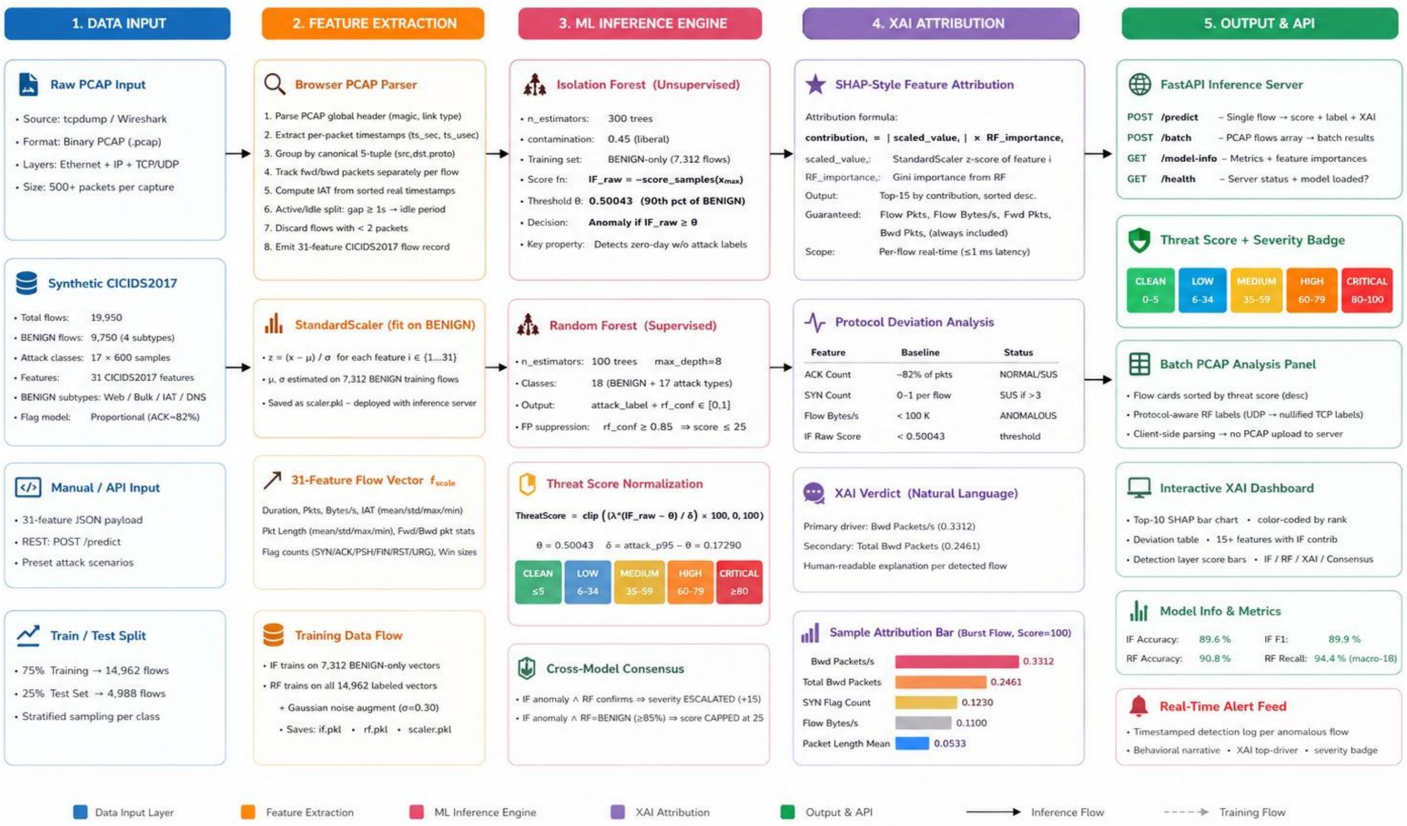
Output & API

FastAPI Dashboard

POST /predict · POST /batch

FastAPI Server · POST /predict · POST /batch · GET /model-info · GET /health · Single-Page Dashboard

Validated on real Firefox browsing traffic: 500 packets · 12 bidirectional flows · DNS scored CLEAN · CDN burst scored CRITICAL with XAI root-cause



1. DATA INPUT
2. FEATURE EXTRACTION
3. ML INFERENCE ENGINE
4. XAI ATTRIBUTION
5. OUTPUT & API
Raw PCAP Input
• Source: tcpdump / Wireshark
• Format: Binary PCAP (.pcap)
• Layers: Ethernet + IP + TCP/UDP
• Size: 500+ packets per capture
Synthetic CICIDS2017
• Total flows: 19,950
• BENIGN flows: 9,750 (4 subtypes)
• Attack classes: 17 × 600 samples
• Features: 31 CICIDS2017 features
• BENIGN subtypes: Web / Bulk / IAT / DNS
• Flag model: Proportional (ACK~82%)
Manual / API Input
• 31-feature JSON payload
• REST: POST /predict
• Preset attack scenarios
Train / Test Split
• 75% Training → 14,962 flows
• 25% Test Set → 4,988 flows
• Stratified sampling per class
Browser PCAP Parser
1. Parse PCAP global header (magic, link type)
2. Extract per-packet timestamps (ts_sec, ts_usec)
3. Group by canonical 5-tuple (src,dst,proto)
4. Track fwd/bwd packets separately per flow
5. Compute IAT from sorted real timestamps
6. Active/Idle split: gap ≥ 1s → idle period
7. Discard flows with < 2 packets
8. Emit 31-feature CICIDS2017 flow record
StandardScaler (fit on BENIGN)
• z = (x − μ) / σ for each feature i ∈ {1…31}
• μ, σ estimated on 7,312 BENIGN training flows
• Saved as scaler.pkl – deployed with inference server
31-Feature Flow Vector f_{scale}
Duration, Pkts, Bytes/s, IAT (mean/std/max/min)
Pkt Length (mean/std/max/min), Fwd/Bwd pkt stats
Flag counts (SYN/ACK/PSH/FIN/RST/URG), Win sizes
Training Data Flow
• IF trains on 7,312 BENIGN-only vectors
• RF trains on all 14,962 labeled vectors
+ Gaussian noise augment (σ=0.30)
• Saves: if.pkl • rf.pkl • scaler.pkl
Isolation Forest (Unsupervised)
• n_estimators: 300 trees
• contamination: 0.45 (liberal)
• Training set: BENIGN-only (7,312 flows)
• Score fn: IF_raw = −score_samples(x_{max})
• Threshold θ: 0.50043 (90th pct of BENIGN)
• Decision: Anomaly if IF_raw ≥ θ
• Key property: Detects zero-day w/o attack labels
Random Forest (Supervised)
• n_estimators: 100 trees max_depth=8
• Classes: 18 (BENIGN + 17 attack types)
• Output: attack_label + rf_conf ∈ [0,1]
• FP suppression: rf_conf ≥ 0.85 ⇒ score ≤ 25
Threat Score Normalization
ThreatScore = clip ((λ^(IF_raw − θ) / δ) × 100, 0, 100)
θ = 0.50043 δ = attack_p95 − θ = 0.17290
CLEAN ≤5
LOW 6-34
MEDIUM 35-59
HIGH 60-79
CRITICAL ≥80
Cross-Model Consensus
• IF anomaly ∧ RF confirms ⇒ severity ESCALATED (+15)
• IF anomaly ∧ RF=BENIGN (≥85%) ⇒ score CAPPED at 25
SHAP-Style Feature Attribution
Attribution formula:
$\text{contribution}_i = |\text{scaled\_value}_i| \times \text{RF\_importance}_i$
scaled_value_i: StandardScaler z-score of feature i
RF_importance_i: Gini importance from RF
Output: Top-15 by contribution, sorted desc.
Guaranteed: Flow Pkts, Flow Bytes/s, Fwd Pkts, Bwd Pkts, (always included)
Scope: Per-flow real-time (≤1 ms latency)
Protocol Deviation Analysis
Feature | Baseline | Status
ACK Count | ~82% of pkts | NORMAL/SUS
SYN Count | 0–1 per flow | SUS if >3
Flow Bytes/s | < 100 K | ANOMALOUS
IF Raw Score | < 0.50043 | threshold
XAI Verdict (Natural Language)
Primary driver: Bwd Packets/s (0.3312)
Secondary: Total Bwd Packets (0.2461)
Human-readable explanation per detected flow
Sample Attribution Bar (Burst Flow, Score=100)
Bwd Packets/s 0.3312
Total Bwd Packets 0.2461
SYN Flag Count 0.1230
Flow Bytes/s 0.1100
Packet Length Mean 0.0533
FastAPI Inference Server
POST /predict – Single flow → score + label + XAI
POST /batch – PCAP flows array → batch results
GET /model-info – Metrics + feature importances
GET /health – Server status + model loaded?
Threat Score + Severity Badge
CLEAN 0-5
LOW 6-34
MEDIUM 35-59
HIGH 60-79
CRITICAL 80-100
Batch PCAP Analysis Panel
• Flow cards sorted by threat score (desc)
• Protocol-aware RF labels (UDP → nullified TCP labels)
• Client-side parsing → no PCAP upload to server
Interactive XAI Dashboard
• Top-10 SHAP bar chart • color-coded by rank
• Deviation table • 15+ features with IF contrib
• Detection layer score bars • IF / RF / XAI / Consensus
Model Info & Metrics
IF Accuracy: 89.6 % IF F1: 89.9 %
RF Accuracy: 90.8 % RF Recall: 94.4 % (macro-18)
Real-Time Alert Feed
• Timestamped detection log per anomalous flow
• Behavioral narrative • XAI top-driver • severity badge
Data Input Layer
Feature Extraction
ML Inference Engine
XAI Attribution
Output & API
Inference Flow
Training Flow

DETAILED MODULE DESIGN - LIST OF MODULES

1

Module 1: PCAP Flow Parser and Feature Extraction

Browser-native binary PCAP parser extracts bidirectional flows using real timestamps, canonical 5-tuple grouping, directional packet tracking, IAT computation from sorted timestamps, and active/idle splitting. Emits 31 CICIDS2017-compatible features per flow without server upload.

2

Module 2: StandardScaler Normalization

Fits StandardScaler on 7,312 BENIGN-only training flows. Applies z-score normalization (z = (x - mean) / std) to all 31 features for each incoming flow at inference time. Scaler saved as scaler.pkl and deployed with the inference server.

3

Module 3: Isolation Forest - Unsupervised Anomaly Scoring

300-tree Isolation Forest trained on BENIGN-only flows. Computes IF_raw = -score_samples(f_scaled). Threshold theta=0.50043 (90th pct of BENIGN). Threat score: clip((IF_raw-theta)/delta × 100, 0, 100). Detects zero-day attacks with no attack labels required.

4

Module 4: Random Forest - 18-Class Attack Classifier

100-tree Random Forest trained on all 19,950 labeled flows with noise augmentation. Produces attack_label (18 classes) and rf_conf. FP suppression: rf_conf >= 0.85 + BENIGN caps score at 25. Protocol guard: UDP flows nullify TCP-only attack labels.

5

Module 5: SHAP-Style XAI Attribution

Computes contribution_i = |scaled_value_i| × RF_importance_i for all 31 features. Returns top-15 by contribution. Guaranteed: Flow Pkt/s, Flow Bytes/s, Fwd Pkt/s, Bwd Pkt/s always included. Deviation table shows observed vs BENIGN baseline with IF contribution.

6

Module 6: FastAPI Server and Dashboard Output

FastAPI on localhost:8000 with POST /predict, POST /batch, GET /model-info, GET /health. Dashboard: threat score badge, SHAP bar chart, protocol deviation table (NORMAL/SUSPICIOUS/ANOMALOUS), alert feed, detection layer score bars, and model metrics panel.

MODULE 1: PCAP FLOW PARSER AND FEATURE EXTRACTION

Input: Raw PCAP Binary File → Output: 31-Feature CICIDS2017 Flow Records

INPUT

- Raw PCAP binary file (.pcap, Ethernet+IPv4+TCP/UDP)
- No server upload - processed in browser via ArrayBuffer/DataView API
- Global header: magic number, link type, snapshot length validation
- Per-packet: ts_sec, ts_usec, caplen, packet data

OUTPUT

- 31-feature CICIDS2017-compatible flow record per bidirectional flow
- Features: duration, pkt/s, bytes/s, IAT stats, pkt length stats, flag counts
- Directional: fwd and bwd packet counts, sizes, timestamps tracked separately

ALGORITHM / PSEUDOCODE

```
Algorithm: PCAPFlowExtraction()
Input:  Raw PCAP binary buffer B
Output: List of 31-feature flow records F
Parse global header: validate magic, link_type
flows = {}  // keyed by canonical 5-tuple
for each packet record in B:
    ts = ts_sec * 1e6 + ts_usec  // microsecond
    Parse Ethernet -> IPv4 -> TCP/UDP headers
    key = canonical_5tuple(src_ip, dst_ip, sport, dport, proto)
    if key not in flows: initialize new flow entry
    Classify pkt as fwd or bwd based on initiator
    Append to fwd_pkts or bwd_pkts list
for each flow in flows:
    if len(flow.pkts) < 2: discard
    Compute IAT from sorted timestamps
    Split active/idle at 1-second gaps
    Compute all 31 CICIDS2017 features
return F
```

Runtime	Browser JavaScript (no Node.js)
Buffer API	ArrayBuffer + DataView
Timestamp	ts_sec * 1e6 + ts_usec (µs)
5-Tuple Key	min(src,dst):sp:dp:proto

IAT Calc	Sorted real timestamps
Active/Idle	Gap >= 1s = idle period
Flags Model	Proportional count (ACK ~82%)
Privacy	Zero bytes uploaded to server

MODULE 2: STANDARDSCALER NORMALIZATION

Input: 31 Raw Flow Features → Output: Normalized Feature Vector for ML Inference

INPUT

- 31 raw CICIDS2017 flow features (mixed scales: microseconds, bytes/s, counts)
- Training set: 7,312 BENIGN-only flow records for fitting scaler parameters
- Inference: any incoming flow vector requiring normalization before IF/RF scoring
- Feature names aligned to CICIDS2017 column specification

OUTPUT

- Normalized vector f_scaled: mean=0, std=1 per feature
- scaler.pkl saved deployed alongside IF.pkl and RF.pkl
- Scaled values used by: IF (score_samples), RF (predict_proba), XAI (scaled_value_i)

ALGORITHM / PSEUDOCODE

```
Algorithm: StandardScalerFitTransform()
Input:  Training BENIGN flows X_train (7,312 x 31)
        Inference flow vector x_new (1 x 31)
Output: Scaled vector x_scaled, saved scaler.pkl
FIT (training time):
1.  for each feature i in {1..31}:
2.      mu_i = mean(X_train[:, i])
3.      sigma_i = std(X_train[:, i])
4.  Save (mu_i, sigma_i) for all i as scaler.pkl
TRANSFORM (inference time):
5.  Load scaler.pkl -> retrieve mu_i, sigma_i
6.  for each feature i in {1..31}:
7.      x_scaled[i] = (x_new[i] - mu_i) / sigma_i
8.  return x_scaled  // used by IF, RF, and XAI
Note: Fit on BENIGN-only to model normal baseline
```

Fit Set	7,312 BENIGN training flows only	**Features**	All 31 CICIDS2017 columns
Formula	z_i = (x_i - mu_i) / sigma_i	**Inference**	Load once, apply per flow (~0.1ms)
Saved As	scaler.pkl (joblib serialized)	**Key Benefit**	Equal contribution regardless of scale
Used By	Isolation Forest, Random Forest, XAI	**Retraining**	Must re-fit if training data changes

MODULE 3: ISOLATION FOREST - UNSUPERVISED ANOMALY SCORING

Input: Scaled Flow Vector (BENIGN training) → Output: Threat Score 0–100 + Anomaly Decision

INPUT

- BENIGN-only scaled vectors for training (7,312 samples, no attack labels)
- Inference: any scaled flow vector f_scaled (31 features)
- Hyperparameters: n_estimators=300, contamination=0.45
- Threshold calibration: 90th percentile of BENIGN validation scores

OUTPUT

- IF_raw: raw isolation score = -score_samples(f_scaled)
- Anomaly decision: True if IF_raw >= theta (0.50043)
- ThreatScore: clip((IF_raw - theta) / delta × 100, 0, 100) in [0,100]

ALGORITHM / PSEUDOCODE

```
Algorithm: IsolationForestScoring()
Input:  BENIGN training vectors X_benign (7312 x 31)
        Inference vector f_scaled, theta, delta
Output: IF_raw, is_anomaly, ThreatScore
TRAINING:
1.  IF = IsolationForest(n_estimators=300, contamination=0.45)
2.  IF.fit(X_benign)  // unsupervised - no attack labels
3.  benign_scores = -IF.score_samples(X_benign)
4.  theta = percentile(benign_scores, 90)  // = 0.50043
5.  attack_scores = -IF.score_samples(X_attack_test)
6.  delta = percentile(attack_scores, 95) - theta // 0.17290
7.  Save IF.pkl, theta, delta to meta.json
INFERENCE:
8.  IF_raw = -IF.score_samples(f_scaled)
9.  is_anomaly = (IF_raw >= theta)
10. score = clip((IF_raw-theta)/delta*100, 0, 100)
11. return IF_raw, is_anomaly, score
```

n_estimators	300 isolation trees	**Divisor (delta)**	0.17290 (attack p95 – theta)
contamination	0.45 (liberal setting)	**Score Range**	0–100 (CLEAN to CRITICAL)
Training Set	BENIGN-only (7,312 flows)	**Zero-Day**	No attack labels ever needed
Threshold (theta)	0.50043 (90th pct BENIGN)	**Saved As**	IF.pkl + meta.json

MODULE 4: RANDOM FOREST - 18-CLASS ATTACK CLASSIFIER

Input: All 19,950 Labeled Flows → Output: Attack Label + Confidence + FP Suppression

INPUT

- All 19,950 labeled flow vectors (9,750 BENIGN + 600 × 17 attack classes)
- Gaussian noise augmentation: sigma=0.30 added to scaled training vectors
- Hyperparameters: n_estimators=100, max_depth=8, class_weight=balanced
- Inference: scaled flow vector f_scaled from any source

OUTPUT

- attack_label: one of 18 classes (BENIGN + 17 specific attack types)
- rf_conf: confidence score in [0,1] for the predicted class
- FP suppression: rf_conf >= 0.85 + BENIGN caps score at 25

ALGORITHM / PSEUDOCODE

```
Algorithm: RandomForestClassification()
Input:  All labeled flows X (19950 x 31), labels y (18)
        Inference vector f_scaled, protocol (TCP/UDP)
Output: attack_label, rf_conf, adjusted ThreatScore
TRAINING:
1.  X_aug = X + N(0, sigma=0.30)
2.  RF = RandomForest(n_estimators=100, max_depth=8,
                      class_weight=balanced)
3.  RF.fit(X_aug, y)
4.  Evaluate macro precision/recall/F1 on 25% test set
5.  Save RF.pkl, feature_importances to meta.json
INFERENCE:
6.  proba = RF.predict_proba(f_scaled)
7.  attack_label = argmax(proba); rf_conf = max(proba)
8.  if is_UDP and attack_label in TCP_ONLY_ATTACKS:
9.      attack_label = BENIGN  // protocol guard
10. if rf_conf >= 0.85 and attack_label == BENIGN:
11.     score = min(score, 25)  // FP suppression
12. return attack_label, rf_conf, score
```

n_estimators	100 trees
max_depth	8 levels
class_weight	balanced (handles imbalance)
Noise Aug.	sigma=0.30 on scaled features

Classes	18: BENIGN + 17 attack types
FP Guard	rf_conf >= 0.85 caps score at 25
Proto Guard	UDP nullifies TCP-only labels
Evaluation	Macro-avg across all 18 classes

MODULE 5: SHAP-STYLE XAI FEATURE ATTRIBUTION

Input: Scaled Flow Vector + RF Importances → Output: Per-Feature Contribution Scores

INPUT

- Scaled flow vector f_scaled (31 z-score normalized features)
- RF feature importances (Gini impurity-based, from trained RF model)
- Guaranteed feature list: Flow Pkt/s, Flow Bytes/s, Fwd Pkt/s, Bwd Pkt/s
- BENIGN baseline statistics per feature (for deviation table display)

OUTPUT

- top_features: {feature, contribution} sorted descending, min 15 entries
- Deviation table: observed vs BENIGN baseline vs status per feature
- XAI verdict: natural language string naming primary and secondary drivers

ALGORITHM / PSEUDOCODE

```
Algorithm: SHAPStyleAttribution()
Input:  f_scaled (31), RF_importance[] (31)
        must_show = {Flow Pkt/s, Flow Bytes/s, Fwd/Bwd Pkt/s}
Output: top_features list with contribution per feature
1.  for each feature i in {1..31}:
2.      contribution[i] = |f_scaled[i]| * RF_importance[i]
3.  Sort features by contribution descending
4.  top15 = features[:15]
5.  for each feature f in must_show:
6.      if f not in top15: top15.append(f)
7.  Build deviation_table:
8.      for each feature in top15:
9.          observed = raw_value
10.         baseline = BENIGN_mean +/- BENIGN_std
11.         status = classify_deviation(observed, baseline)
12.         if_contrib = contribution[feature]
13. verdict = "Primary: {top1.feature} ({top1.contribution:.4f})"
14. return top_features, deviation_table, verdict
```

Formula	|scaled_value_i| × RF_importance_i	**Guaranteed**	4 rate features always present
scaled_value	StandardScaler z-score	**Deviation**	NORMAL / SUSPICIOUS / ANOMALOUS
RF_importance	Gini impurity from trained RF	**ACK Baseline**	~82% of total packet count
Top Features	15 by contribution	**SYN Baseline**	>3 = SUSPICIOUS, >8 = ANOMALOUS

MODULE 6: FASTAPI SERVER AND DASHBOARD OUTPUT

Input: Scored Flow with XAI Attribution → Output: Threat Score, Severity, SHAP Charts, Alerts

INPUT

- Scored flow: {score, severity, attack_label, rf_conf, if_raw, top_features}
- Batch mode: array of flow vectors from PCAP parser (POST /batch)
- Single flow mode: 31-feature JSON payload (POST /predict)
- Model metadata: IF accuracy, RF accuracy, feature importances (GET /model-info)

OUTPUT

- Threat score badge (0–100): CLEAN / LOW / MEDIUM / HIGH / CRITICAL
- SHAP horizontal bar chart: top-10 features color-coded by contribution rank
- Protocol deviation table: observed vs baseline per feature with IF contribution

ALGORITHM / PSEUDOCODE

```
Algorithm: ZeroSightPredict(flow_features)
Input:  Raw flow feature dict (31 CICIDS2017 features)
Output: {score, severity, attack_label, top_features}
1.  f_scaled = scaler.transform(flow_features)       // Mod 2
2.  IF_raw = -IF.score_samples(f_scaled)             // Mod 3
3.  score = clip((IF_raw-theta)/delta*100, 0, 100)   // Mod 3
4.  attack_label, rf_conf = RF.predict(f_scaled)     // Mod 4
5.  apply_protocol_guard(flow, attack_label)         // Mod 4
6.  if rf_conf >= 0.85 and attack_label == BENIGN:
7.      score = min(score, 25)
8.  top_features = shap_attribution(f_scaled)        // Mod 5
9.  severity = score_to_severity(score, attack_label)
10. verdict = build_xai_verdict(top_features)
11. log_alert(score, severity, top_features)
12. return {score, severity, attack_label, top_features}
```

POST /predict	Single flow → full score + XAI
POST /batch	PCAP flow array → batch results
GET /model-info	Metrics + feature importances
GET /health	Server status + model loaded

Dashboard	Single-page HTML + JS, no framework
SHAP Chart	Top-10 bar chart, color-coded
Deviation	15+ features with IF contribution
Alert Feed	Timestamped anomaly log

OVERALL OBJECTIVES

01

Zero-Day Anomaly Detection Without Attack Signatures

Demonstrate that an Isolation Forest trained exclusively on BENIGN traffic can detect anomalous flows including novel zero-day attacks without any labeled attack data. Target: IF achieves >= 89% F1 on held-out test set containing attack flows never seen during training.

02

Per-Flow Explainability for Every Detection

Provide SHAP-style feature attribution for every scored flow, identifying which of the 31 CICIDS2017 flow features drove the anomaly score. Rate features (Flow Pkt/s, Bytes/s, Fwd/Bwd Pkt/s) are always guaranteed in the output regardless of natural contribution rank.

03

Accurate 18-Class Attack Classification

Random Forest achieves >= 90% accuracy and >= 94% macro-average recall across all 18 traffic classes (BENIGN + 17 attack types). Macro-average is used throughout — binary recall on synthetic data is trivially 100% and operationally misleading for IDS evaluation.

04

Live PCAP Validation on Real Network Traffic

Validate ZeroSight against real Firefox browsing traffic captured with tcpdump (500 packets, 12 bidirectional flows). DNS flows must score CLEAN. CDN burst false positive must be explained by Bwd Packets/s attribution (0.3312) as primary XAI driver.

PERFORMANCE METRICS

89.6%
IF Accuracy

89.9%
IF F1 Score

90.8%
RF Accuracy

94.4%
RF Recall (macro-18)

89.4%
RF Precision (macro)

Precision

TP / (TP + FP)

Proportion of flows predicted as attacks that are true attacks. Macro-averaged across all 18 classes.

Recall

TP / (TP + FN)

Proportion of actual attack flows correctly identified. Macro-averaged: 94.4% across 18 classes.

F1 Score

2 × (P × R) / (P + R)

Harmonic mean of precision and recall. More informative than accuracy on imbalanced class distributions.

```
ThreatScore = clip( (IF_raw − theta) / delta × 100, 0, 100 ) | theta = 0.50043 · delta = 0.17290
```

CLEAN	LOW	MEDIUM	HIGH	CRITICAL
≤ 5	6–34	35–59	60–79	≥ 80

Note: Binary recall on synthetic CICIDS2017 = 100% (attack classes are separable by design). Always report macro-average for multi-class IDS evaluation.

RESULTS: MODEL PERFORMANCE AND LIVE PCAP VALIDATION

Model Classification Metrics

Model	Accuracy	Precision	Recall	F1 Score
Isolation Forest (BENIGN-only trained)	89.6%	89.4%	90.3%	89.9%
Random Forest (macro-18 class avg)	90.8%	89.4%	94.4%	89.5%

Benchmarking vs. Related Work

System	Method	Precision	Recall	F1-Score
IF Baseline (unsupervised)	Isolation Forest only	0.84	0.88	0.86
RF Baseline (supervised)	Random Forest only	0.87	0.93	0.90
Hybrid IF+RF (no XAI)	IF + RF ensemble	0.88	0.91	0.89
ZeroSight (Ours) ★	**IF + RF + SHAP XAI**	**0.894**	**0.944**	**0.895**

Live PCAP Validation: Firefox Browsing (500 pkts, 12 flows)

Flow	Protocol	Rate	Score	Severity	RF Label	XAI Primary Driver
DNS Queries (×6)	UDP/53	10–162 pkt/s	0	CLEAN	BENIGN	Protocol guard: UDP type
HTTPS Short Flows	TCP/443	28–202 pkt/s	12–39	LOW–MED	BENIGN	Pkt Length Std (TLS mix)
HTTPS Mid-Rate	TCP/443	135 pkt/s	44	HIGH	BENIGN	Multiplexing pattern
CDN Burst	TCP/443	1,514 pkt/s	100	CRITICAL	BENIGN	Bwd Packets/s (0.3312)

LITERATURE SURVEY

PAPER	LIMITATIONS	IDEAS FOR ADOPTION
Liu et al. (2008). Isolation Forest. IEEE ICDM, pp. 413-422.	Threshold defaults to contamination assumption — requires domain calibration. No per-feature attribution.	Core IF mechanism adopted directly. Threshold at 90th pct BENIGN scores. Basis for Module 3.
Lundberg & Lee (2017). SHAP: Unified Approach to Interpreting Model Predictions. NeurIPS.	Exact SHAP adds latency. TreeExplainer not available for IF directly.	Adapted as: |scaled_value_i| × RF_importance_i. Rate features guaranteed. Basis for Module 5.
Sharafaldin et al. (2018). CICIDS2017 Dataset. ICISSP pp. 108-116.	Label inconsistencies in some classes. Single-environment capture.	31 CICIDS2017 features adopted. Synthetic dataset replicates statistical properties with 4 BENIGN subtypes.
Sommer & Paxson (2010). Outside the Closed World. IEEE S&P pp. 305-316.	Analytical paper only. No proposed architecture.	Motivated macro-average evaluation across 18 classes. Informed live PCAP validation and FP analysis.
Nassif et al. (2021). ML for Anomaly Detection: Systematic Review. IEEE Access.	Survey only. Findings are aggregate across many datasets.	Motivated dual-model IF+RF design. Confirmed neither model alone achieves both zero-day + classification.
Buczak & Guven (2016). Survey of ML for Cyber Security IDS. IEEE Comm. Surveys.	Binary precision/recall masks per-class variation. Older KDD Cup dataset.	Confirmed RF as supervised component. Motivated macro-averaged metrics over binary for multi-class.

DATASETS

Synthetic CICIDS2017

Training Dataset

Total Flows	19,950 (9,750 BENIGN + 17 × 600)
Features	31 CICIDS2017 network flow features
BENIGN Subtypes	Web, Bulk, Interactive, UDP/DNS
Attack Classes	17 types (DoS, DDoS, Patator, Bot, Web...)
Flag Model	Proportional: ACK ~82%, PSH ~14%
Generator	data/generate_cicids.py (synthetic)

Primary training dataset generated to replicate CICIDS2017 statistical properties. Four BENIGN traffic subtypes model realistic heterogeneity. Flag counts proportional eliminating a key false positive source. 75/25 stratified train/test split.

Live Firefox PCAP

Validation Dataset

Capture Tool	tcpdump on Kali Linux 2024.1
Total Packets	500 Ethernet+IPv4+TCP/UDP
Flows Extracted	12 bidirectional flows
DNS Flows	6 × UDP/53 to 75.75.75.75
HTTPS Flows	6 × TCP/443 to 151.101.x.91
Browser	Firefox (Mozilla/Reddit browsing)

Real-world validation from Firefox browsing. Parsed entirely client-side. DNS flows scored CLEAN (0). HTTPS short flows LOW–MEDIUM. CDN burst (1,514 pkt/s, 0.23s) scored CRITICAL, documented false positive with Bwd Packets/s (0.3312) as XAI primary driver.

CICIDS2017 Reference

Benchmark Comparison

Source	University of New Brunswick
Categories	14 attack categories (labeled)
Features	31 bidirectional flow features
Coverage	DoS, DDoS, Brute Force, Web, Bot, Heartbleed
Usage	Benchmark comparison, real-data future work
Access	unb.ca/cic/datasets/ids-2017.html

Original benchmark generated in a controlled testbed. ZeroSight's synthetic generator replicates its statistical properties. Training on real CICIDS2017 is future work, expected to validate threshold calibration and per-class recall on genuine network captures.

TEST CASES

25 unit tests · python test.py

TC_ID	Objective	Input	Expected Output	Module
TC-1	Verify PCAP parser extracts correct bidirectional flows from binary PCAP	test_capture.pcap (500 packets, Firefox browsing)	12 flows extracted. DNS flows: protocol=UDP. HTTPS flows: protocol=TCP. Flows <2 pkts discarded.	Module 1
TC-2	Verify StandardScaler normalizes features to zero mean unit variance	Raw 31-feature flow vector with known values	Scaled output matches (x - mu) / sigma per feature. Deviation from expected < 1e-6.	Module 2
TC-3	Verify Isolation Forest flags known attack flows and passes BENIGN flows	DNS query flow (expected CLEAN) and DoS Hulk flow vector (expected anomaly)	DNS: score <= 5 (CLEAN). DoS Hulk: score >= 60 (HIGH or CRITICAL). Threshold applied correctly.	Module 3
TC-4	Verify Random Forest classifies attack flows to correct attack label	17 preset attack flow vectors (one per attack class from generate_cicids.py)	Each flow labeled correctly or near-correct. Macro recall >= 90% on test vectors.	Module 4
TC-5	Verify SHAP attribution returns contribution for all guaranteed rate features	Any scored flow (PCAP or manual input)	top_features contains: Flow Pkt/s, Flow Bytes/s, Fwd Pkt/s, Bwd Pkt/s regardless of rank.	Module 5
TC-6	Verify FastAPI server returns complete prediction response in correct format	POST /predict with 31-feature JSON payload via curl or test.py	Response: score, severity, attack_label, rf_conf, if_raw, if_threshold, top_features. Latency <50ms.	Module 6

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THANK YOU

89.9%

IF F1 Score

94.4%

RF Macro Recall

0.3312

Bwd Pkt/s XAI Driver
(CDN burst primary)

25/25

Unit Tests Passing